

USEPA
DMR-QA 34

NPDES PERMITTEE
DATA REPORT FORM



ERA

A Waters Company

USEPA NPDES MI0020257

Permit #:

Permit Ext:

WATER RESOURCES
DIVISION

DEC 10 2014

Permittee Name: Cadillac WWTP

CADILLAC DISTRICT

Facility Address: 1121 Plett Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

For DMRQA-34, conducted in 2014, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 11, 2014

Yes ☒ No ☐

Received a Graded Report by
August 1, 2014

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative

(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Amy Vail

Title: Lab Manager

Signature:

Date Signed: 8/11/2014

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

AUG 18 2014

Please make corrections in boxes above

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WATER RESOURCES DIVISION
GRAND RAPIDS

United States

ENVIRONMENTAL PROTECTION AGENCY

Laboratory Performance Evaluation
Laboratory DMR-QA Evaluation Study 34

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			CHEM	MICRO	TOX		
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G	☒
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒

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* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMRQA-34, Frequently Asked Questions page

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist**DMRQA Study 34**

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃				
pH pH				
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P				
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P				
Nitrite Nitrite as N				
Demand BOD CBOD COD TOC				

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Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 34

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Oil & Grease Concentrate				
n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease				
n-Hexane Extractable Material(O&G) (Grav)				
n-Hexane Extractable Material(O&G) (IR)				
Trace Metals				
Aluminum				
Antimony				
Arsenic				
Barium				
Beryllium				
Boron				
Cadmium				
Chromium				
Cobalt				
Copper				
Iron				
Lead				
Manganese				
Molybdenum				
Nickel				
Selenium				
Silver				
Strontium				
Thallium				
Vanadium				
Zinc				
Mercury				
Mercury				
Low-Level Mercury				
Low Level Mercury EPA 1631E	☐	☒	☐	☐
Hexavalent Chromium				
Hexavalent Chromium				
Turbidity				
Turbidity				
Total Cyanide				
Cyanide, total Amenable Cyanide				
Total Phenolics (4-AAP)				
Phenolics, total				

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Permittee Name: Cadillac WWTP

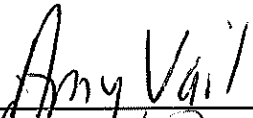
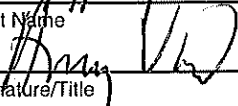
Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 34

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				


 Print Name

 Signature/Title
 Lab Manager

8/11/14
 Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 34

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids SM 2540 D-1997 Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☐	☒	☐
pH pH SM 4500-H+ B-2011	☐	☒	☐	☐
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N SM 4500-NH ₃ D-2011 Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P	☐	☐	☒	☐
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P SM 4500-P E-2011	☐	☒	☐	☐
Nitrite Nitrite as N				

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Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 34

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Demand BOD CBOD SM 5210 B-2011 COD TOC	☐	☒	☐	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease n-Hexane Extractable Material(O&G) (Grav) n-Hexane Extractable Material(O&G) (IR)				
Trace Metals Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc		see WS-211 Report for metals (attached) RW 8/11/14		
Mercury Mercury				
Low-Level Mercury Low Level Mercury				
Hexavalent Chromium Hexavalent Chromium				
Turbidity Turbidity				

Permittee Name: Cadillac WWTP

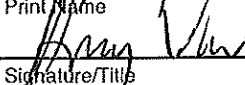
Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 34

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
<u>Total Cyanide</u>				
Cyanide, total				
Amenable Cyanide				
<u>Total Phenolics (4-AAP)</u>				
Phenolics, total				
<u>Total Residual Chlorine</u>				
Total Residual Chlorine				
<u>Low-Level Total Residual Chlorine</u>				
Low Level Total Residual Chlorine				
<u>WasteWatR™ Coliform MicrobE™</u>				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
SM 9222 D (m-FC)-1997				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				


 Print Name
 SH Lab Manager
 Signature/Title
 8/11/14

8/11/14
 Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMR-QA Study 34

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
Fathead minnow (Test Code 13)						
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2000.0	LC50	☐	☒	☐	☐
Fathead minnow (Test Code 14)						
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50				
Fathead minnow (Test Code 15)						
0808	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	IC25 (ON) Growth	☐	☒	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC (ON) Growth	☐	☒	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC Survival	☐	☒	☐	☐
Fathead minnow (Test Code 16)						
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth				
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth				
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Ceriodaphnia dubia (Test Code 19)						
0764	48Hr., Acute Renewal, 25° C, MHSF EPA 2002	LC50	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 20)						
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50				
Ceriodaphnia dubia (Test Code 21)						
0767	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	IC25 Reproduction	☐	☒	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Reproduction	☐	☒	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Survival	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 22)						
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction				
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction				
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Daphnia magna (Test Code 32)						
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2021	LC50	☐	☒	☐	☐

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Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMR-QA Study 34

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
<u>Daphnia pulex (Test Code 38)</u>						
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				
<u>Mysid (Test Code 42)</u>						
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Mysid (Test Code 43)</u>						
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0818	7-day Short term Chronic, Daily Renewal, 40 FSW					
0799	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Inland silverside (Test Code 44)</u>						
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Inland Silverside (Test Code 45)</u>						
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0826	7-day Short term Chronic, Daily Renewal, 40 FSW					
0824	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Sheepshead minnow (Test Code 46)</u>						
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Sheepshead minnow (Test Code 47)</u>						
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0822	7-day Short term Chronic, Daily Renewal, 40 FSW					
0805	7-day Short term Chronic, Daily Renewal, 40 FSW					

Print Name

Signature/Title

Date

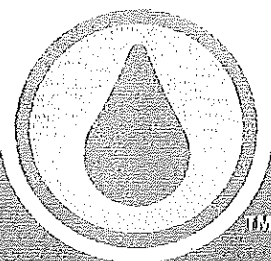
Use a separate checklist for EACH lab used



A Waters Company

Amy Jean Vail
City of Cadillac WWTP Lab
200 N. Lake St.
Laboratory
Cadillac, MI 49601

WS-211



Final Report

WatR™ Supply Proficiency Testing

WatR™ Supply Study

Open Date: 02/10/14

Close Date: 03/27/14

Report Issued Date: 03/31/14



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2009 TNI Evaluation Report

Study: WS-211

ERA Customer Number: C374701

Laboratory Name: City of Cadillac WWTP
Lab

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Inorganic Residues





WS-211 2009 TNI Evaluation Final Complete Report

A Waters Company

Amy Jean Vail
Laboratory Manager
City of Cadillac WWTP Lab
200 N. Lake St.
Laboratory
Cadillac, MI 49601
(231) 775-2368

EPA ID:
ERA Customer Number:
Report Issued:
Study Dates:

Not Reported
C374701
03/31/14
02/10/14 - 03/27/14

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
WS Metals (cat# 590)												
1000	Aluminum	µg/L		690	586 - 794	Not Reported				688	32.6	
1005	Antimony	µg/L		24.3	17.0 - 31.6	Not Reported				24.5	1.79	
1010	Arsenic	µg/L		25.5	17.8 - 33.2	Not Reported				25.5	2.01	
1015	Barium	µg/L		1260	1070 - 1450	Not Reported				1240	56.3	
1020	Beryllium	µg/L		11.3	9.60 - 13.0	Not Reported				11.7	0.729	
1025	Boron	µg/L		1700	1440 - 1960	Not Reported				1700	104	
1030	Cadmium	µg/L	22.7	23.3	18.6 - 28.0	Acceptable	SM3113B online	2/17/2014	-0.506	23.3	1.19	
1040	Chromium	µg/L	83.7	83.7	71.1 - 96.3	Acceptable	SM3113B online	3/18/2014	0.213	82.9	3.90	
1055	Copper	µg/L	1284	1260	1130 - 1390	Acceptable	SM3113B online	3/25/2014	0.525	1250	57.3	
1070	Iron	µg/L		1190	1010 - 1370	Not Reported				1200	50.1	
1075	Lead	µg/L	43.7	42.3	29.6 - 55.0	Acceptable	SM3113B online	3/14/2014	0.456	42.4	2.90	
1090	Manganese	µg/L		77.1	65.5 - 88.7	Not Reported				78.5	3.99	
1100	Molybdenum	µg/L	100.0	107	91.0 - 123	Acceptable	SM3113B online	3/17/2014	-0.768	105	7.12	
1105	Nickel	µg/L	28	26.0	22.1 - 29.9	Acceptable	SM3113B online	3/26/2014	1.47	25.7	1.59	
1140	Selenium	µg/L	75.7	73.0	58.4 - 87.6	Acceptable	SM3113B online	3/20/2014	0.887	71.8	4.35	
1150	Silver	µg/L	56.8	56.8	39.8 - 73.8	Acceptable	SM3113B online	3/19/2014	-0.0947	57.1	2.95	
1165	Thallium	µg/L		7.32	5.12 - 9.52	Not Reported				7.22	0.469	
1185	Vanadium	µg/L		762	648 - 876	Not Reported				753	37.8	
1190	Zinc	µg/L	1248	1130	960 - 1300	Acceptable	SM3113B online	3/25/2014	1.83	1140	58.3	
WS Turbidity (cat# 592)												
2055	Turbidity	NTU	3.31	3.36	2.83 - 3.85	Acceptable	SM2130B online	3/26/2014	-0.387	3.39	0.219	



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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DMR-QA 34 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/25/14
Study Dates: 03/21/14 - 07/11/14

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580)											
1960	Total Suspended Solids	mg/L	Not Acceptable	33	44.6	34.0 - 51.5	SM 2540 D-1997 1997	41.4	3.02	MI000082	DMRQA34
DMRQA pH (cat# 577)											
1900	pH	S.U.	Acceptable	7.84	7.84	7.64 - 8.04	SM 4500-H+ B-2011 2011	7.86	0.0722	MI000082	DMRQA34
DMRQA Simple Nutrients (cat# 584)											
1515	Ammonia as N	mg/L	Not Acceptable	9.13	7.34	5.78 - 8.90	SM 4500-NH3 D-2011 2011	7.39	0.598	MI000082	DMRQA34
DMRQA Complex Nutrients (cat# 579)											
1910	Total phosphorus as P	mg/L	Acceptable	1.86	1.69	1.35 - 2.02	SM 4500-P E-2011 2011	1.69	0.115	MI000082	DMRQA34
DMRQA Demand (cat# 578)											
1555	CBOD	mg/L	Acceptable	37.5	57.1	25.7 - 88.5	SM 5210 B-2011 2011	61.4	13.5	MI000082	DMRQA34
DMRQA Low-Level Mercury (cat# 896)											
1095	Low Level Mercury	ng/L	Acceptable	86.2	78.4	60.0 - 95.8	EPA 1631E 2002	77.6	10.1	MI000041	DMRQA34



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 34
Page 2 of 4





A Waters Company

Study: **DMR-QA 34**

NPDES Permit #: **MI0020257**

Laboratory Name: **Cadillac WWTP**

Microbiology Results





DMR-QA 34 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Platt Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/25/14
Study Dates: 03/21/14 - 07/11/14

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
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QA WasteWater™ Coliform Microbe™ (cat# 576)

2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	240	1510	15.0 - 3040	SM 9222 D (m-FC)-1397	216	505	MI00082	DMR-QA34
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Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com

Study #: DMR-QA 34
Page 4 of 4





200 North Lake Street • Cadillac, Michigan 49601
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www.cadillac-mi.net

9/3/14

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Chris Veldkamp
MDEQ Water Resources
State of Michigan Building
350 Ottawa Ave. NW Unit 10
Grand Rapids, MI 49503

SEP 05 2014

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Ms. Veldkamp,

This letter and its supporting documents are to serve as the City of Cadillac WWTP's (MI0000257) corrective actions for unacceptable results from this year's DMR-QA Study 34. We had an "unacceptable" result for total suspended solids and ammonia in this year's testing. We do our total suspended solids and ammonia testing in-house.

Our total suspended solids results were reported as 33.0 mg/L with an actual value of 44.6 mg/L and an acceptable range of 34.0-51.5 mg/L. The 33.0 mg/L we obtained was from averaging two samples one which was 28.0 mg/L and the other was 38.0 mg/L (which was acceptable). The 28.0 mg/L result was from a 50 mL sample and the 38.0 mg/L result was from a 100 mL sample. I believe the lower sample volume resulted in the less accurate reading. We ran our corrective action sample (WP234) using volumes of 200 and 250 mL volumes and passed it. I have included a copy of the report with this letter.

To make sure we do not have future issues with our total suspended solids analysis, our lab has taken the following actions: We will use higher volumes that allow the samples to filter in sufficient time and give us more accurate results. In the future we will use higher volumes for our NPDES and DMR-QA tests.

Our ammonia results were reported as 9.13 mg/L with an actual value of 7.34 mg/L and an acceptable range of 7.64-8.04 mg/L. I believe the reason for this elevated result was due to our ammonia probe. Soon after the testing period we began to notice that the values near the end of a daily run seemed to trend upwards. Our calibrations and initial check were in the acceptable ranges. The DMR-QA34 sample was ran as the final sample which may have caused the elevated result. We have replaced the probe and have ran a corrective action sample (WP234) and passed it.

To make sure we do not have future issues with ammonia results we did replace the probe when the problem was noticed. We have begun to do a check at the end of the run to make sure the values are still acceptable and not trending upwards/downwards as can happen when probes are going bad.

If you have any questions or concerns, please feel free to contact me at (231)775-2368 or lab@cadillac-mi.net.

Thank you,

Amy Vail
City of Cadillac
Lab Manager



A Waters Company

Amy Vail
Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601
USA

WP-234 *Final Report*

WatR™ Pollution Proficiency Testing

WatR™ Pollution Study

Open Date: 07/14/14

Close Date: 08/28/14

Report Issued Date: 09/02/14



A Waters Company

September 2, 2014

Amy Vail
Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

Enclosed is your final report for ERA's WP-234 WatR™ Pollution Proficiency Testing (PT) study. Your final report includes an evaluation of all results submitted by your laboratory to ERA.

Data Evaluation Protocols: All analytes in ERA's WP-234 WatR™ Pollution Proficiency Testing study have been evaluated using the following tiered approach. If the analyte is listed in the current TNI Fields of Proficiency Testing (FoPT) tables, the evaluation was completed by comparing the reported result to the acceptance limits generated using the criteria contained in the current TNI FoPT tables. If the analyte is not included in the TNI FoPT tables, the reported result has been evaluated using the procedures outlined in ERA's Standard Operating Procedure for the Generation of Performance Acceptance Limits (SOP 0260).

Corrective Action Help: As part of your accreditation(s), you may be required to identify the root cause of any "Not Acceptable" results, implement the necessary corrective actions, and then satisfy your PT requirements by participating in a Supplemental (Quik™ Response) or future ERA PT study. ERA's technical staff is available to help your laboratory resolve any technical issues that may be impairing your PT performance and possibly affecting your routine data quality. Our laboratory and technical staff have many years of collective experience in performing the full range of environmental analyses. As part of our technical support, ERA offers QC samples that can be useful in helping you work through your technical issues.

At the request of the TNI Accreditation Council, we have included a Laboratory Exception Report that includes a list of all analytes reported with less than qualifiers when the assigned value was greater than "0." In addition, because we have received many requests from laboratories, this report also includes a list of all analytes with "Not Acceptable" evaluations.

Some states have elected not to convert to the 2009 TNI Standards at this time. If you have released your results to a state that has retained the 2003 NELAC Evaluation Criteria, your final report will include a section that evaluates the results according to the 2003 Standard in addition to the 2009 TNI Standards.

Thank you for your participation in ERA's WP-234 WatR™ Pollution Proficiency Testing study. If you have any questions, please contact our Proficiency Testing Department at 1-800-372-0122.

Sincerely,

Kristina Sanchez
Quality Officer

attachments



A Waters Company

Report Recipient	Contact/Phone Number	Reporting Type	Evaluation Type
Michigan (WP)	George Krisztian / 517-335-8812	All Analytes	2009 TNI



A Waters Company

WP-234 Definitions & Study Discussion

Study Dates: 07/14/14 - 08/28/14

Report Issued: 09/02/14

WP Study Definitions

The Reported Value is the value that the laboratory reported to ERA.

The ERA Assigned Values are compliant with the most current TNI Fields of Proficiency Testing (FoPT) tables. A parameter not added to the standard is given an Assigned Value of "< PTRL" per the guidelines contained in the 2009 TNI Standards. The assigned values are directly traceable to the commercially prepared starting materials used to manufacture the PT standards.

The Acceptance Limits are established per the criteria contained in the most current USEPA/NELAC FoPT tables, or ERA's SOP for the Generation of Performance Acceptance Limits™ as applicable.

The Performance Evaluation:

Acceptable	= Reported Value falls within the Acceptance Limits.
Not Acceptable	= Reported Value falls outside the Acceptance Limits.
No Evaluation	= Reported Value cannot be evaluated.
Not Reported	= No Value reported.

The Method Description is the method the laboratory reported to ERA.

WP Study Discussion

ERA's WP-234 WatR™Pollution Proficiency Testing study has been reviewed by ERA senior management and certified compliant with the requirements of the 2009 TNI PT Standard and the criteria contained in the most current TNI Fields of Proficiency Testing (FoPT) tables.

ERA's WP-234 WatR™Pollution study standards were examined for any anomalies. A full review of all homogeneity, stability and accuracy verification data was completed. All analytical verification data for all analytes met the acceptance criteria contained in the 2009 TNI PT Standard and the criteria contained in the most current TNI FoPT tables.

The data submitted by participating laboratories was also examined for study anomalies. There were no anomalies observed during the statistical review of the data.

ERA's WP-234 WatR™Pollution study reports shall not be reproduced except in their entirety and not without the permission of the participating laboratories. The report must not be used by the participating laboratories to claim product endorsement by any agency of the U. S. government.

The data contained herein are confidential and intended for your use only.

If you have any questions or concerns regarding your assessment in ERA's WatR™Pollution Proficiency Testing program, please contact our Proficiency Testing Department at 1-800-372-0122.



Study # : WP-234





A Waters Company

W.P.-234 Laboratory Exception Report

Amy Vail
Lab Manager
Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601
(231) 775-2368

EPA ID:
ERA Customer Num:
Report Issued:
Study Dates:

2009 TNI Evaluation Checks

There are no values reported with < where the assigned value was greater than 0.

2009 TNI Not Acceptable Evaluations

There were no Not Acceptable evaluations for this study.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



A Waters Company

Final Report Results For Laboratory Cadillac WWTP



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2009 TNI Evaluation Report

Study: **WP-234**

ERA Customer Number: **C454401**

Laboratory Name: **Cadillac WWTP**

Inorganic Results



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A Waters Company

WP-234 2009 TNI Evaluation Final Complete

Amy Vail
Lab Manager
Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601
(231) 775-2368

EPA ID:
ERA Customer Number
Report Issued:
Study Dates:

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score
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NP Hardness (cat# 580)

1960	Total Suspended Solids	mg/L	27.1	34.7	25.3 - 41.0	Acceptable	SM 2540 D-2011 2011	8/12/2014	-1.91
1035	Calcium	mg/L		30.1	25.6 - 34.6	Not Reported			
1085	Magnesium	mg/L		16.0	13.6 - 18.4	Not Reported			
1550	Calcium Hardness as CaCO ₃	mg/L		75.3	64.0 - 86.6	Not Reported			
1755	Total Hardness as CaCO ₃	mg/L		141	120 - 162	Not Reported			

NP Simple Nutrients (cat# 584)

1515	Ammonia as N	mg/L	4.69	4.76	3.67 - 5.89	Acceptable	SM 4500-NH3 D-2011 2011	8/11/2014	-0.246
1820	Nitrate + Nitrite as N	mg/L		17.7	14.8 - 20.4	Not Reported			
1810	Nitrate as N	mg/L		17.7	14.8 - 20.5	Not Reported			
1870	ortho-Phosphate as P	mg/L		4.66	3.96 - 5.36	Not Reported			



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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CERTIFICATE OF EXCELLENCE

In recognition of the quality of your laboratory in proficiency testing

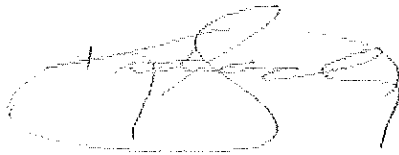
WP-234

Cadillac WWTP

is issued this certificate of achievement by ERA. This laboratory has been recognized as a Laboratory achieving 100% acceptable data in this study which included 1098 participating laboratories. This demonstration of the superior quality of the laboratory in evaluation of the standards list

Hardness

Simple Nutrients



Kristina Sanchez
Quality Officer